

5) OUMV conjecture

for any constant $\epsilon > 0$, even using arbitrary polynomial ~~pre~~ preprocessing time, there is no $O(n^{3-\epsilon})$ time algorithm that solves OUMV with an error probability of at most $1/3$.

★ OUMV problem

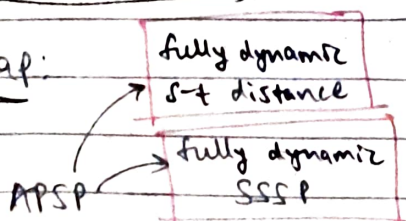
$\Rightarrow$ Given a boolean $n \times n$ matrix M , and an online sequence of n pairs of vectors (u_i, v_i) , after each pair arrives, report $u_i^T M v_i$.

Remark: Any algorithm that solves OMV $\Rightarrow$ can be used for OUMV.
so OMV conjecture $\Rightarrow$ OUMV conjecture

OUMV $\rightarrow$ OMV

8th May, 2026

Recap:



SETH $\rightarrow$

fully dynamic
#SSP

OMV $\rightarrow$

fully dynamic
st reachability

BMM $\rightarrow$

Inc. unweighted SSSP

Dec. unweighted SSSP

Decremental
triangle detection

* Adaptive Updates

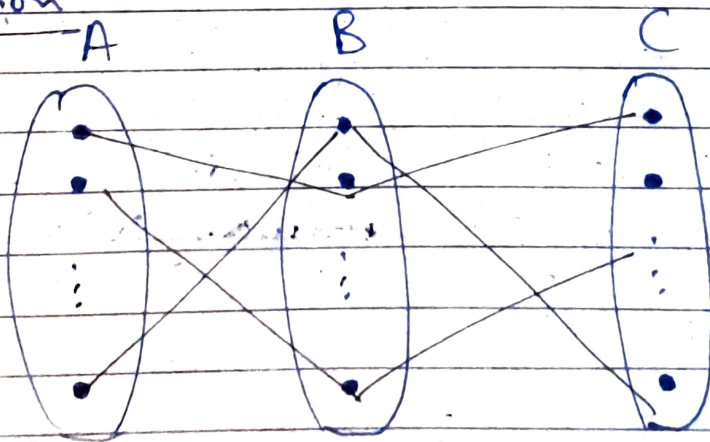
we showed a reduction from BMM to incremental triangle detection problem.

① Next we establish a reduction to the incremental component max clique problem.

Given a graph G , maintain a max clique under deletion of edges for each component.

[a max clique is largest set of vertices where all vertices are connected to each other directly]

* construction



Add edges $(A[i], B[j])$ if $X[i][j] = 1$

Add edges $(B[i], C[j])$ if $Y[i][j] = 1$

* Algorithm

for each component

if Max clique size = 1 $\rightarrow$ ignore

if Max clique size = 3 $\rightarrow (A[i], B[j], C[k])$

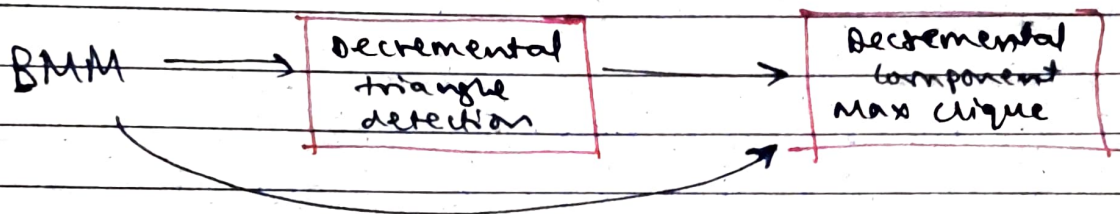
triangle, so $M[i][k] = 1$ and delete $(A[i], C[k])$

If Max clique size = 2

then if $(A[i], z[k]) \rightarrow M[i][k] \geq 0$,
delete $(A[i], c[k])$

if $(A[i], B[j])$ or
 $(B[j], c[k]) \rightarrow$ delete the edge

Hence, we end up computing BMM using a Decremental Component Max Clique Algorithm thus we established:



②

We can further show that BMM can be reduced to Fully dynamic maximal clique.

Given a graph G , maintain a maximal clique under edge ~~additions~~ insertions and deletions

* construction of Algorithm

If Max clique reported is singleton, then we have a problem. So in such a case, we add edges from this vertex to all other vertices. Hence, now max clique will be:

$$\text{maxClique}(G/\{s\}) \cup \{s\}$$

this allows us to perform the same operation as discarding s from G and computing max clique of $G/\{s\}$.

Let's say we perform such operation k times.

If max clique size = $k+1$:

repeat expansion $k+1$

If max clique size = $k+2$:

then if $(A(i), C(k)) \rightarrow M[i][k] = 0$

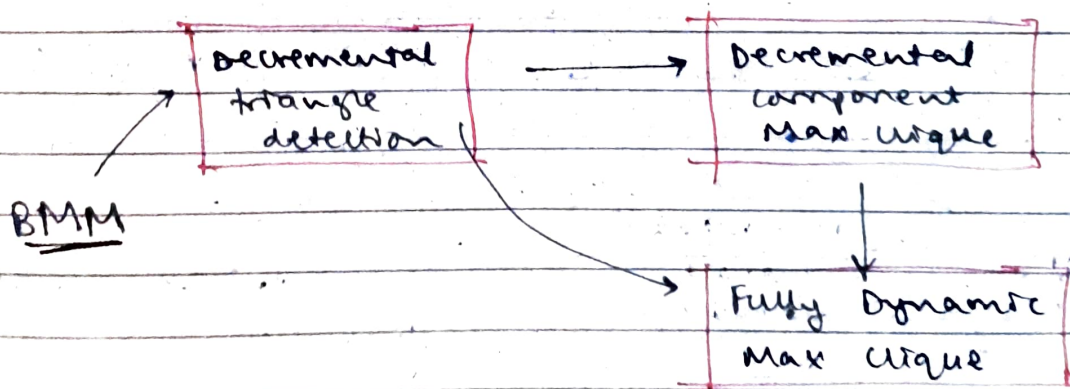
delete the edge

If max clique size = $k+3$:

then $M[i][k] = 1$

delete $(A(i), C(k))$

hence this establishes:



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Another interesting dual between maximal ~~clique~~ clique and maximal independent set is due to the following construction

Given a graph, compute ~~the~~ a maximal set of vertices such that $\nexists$ no edge between any two vertices in the set

the dual exists due to the result:

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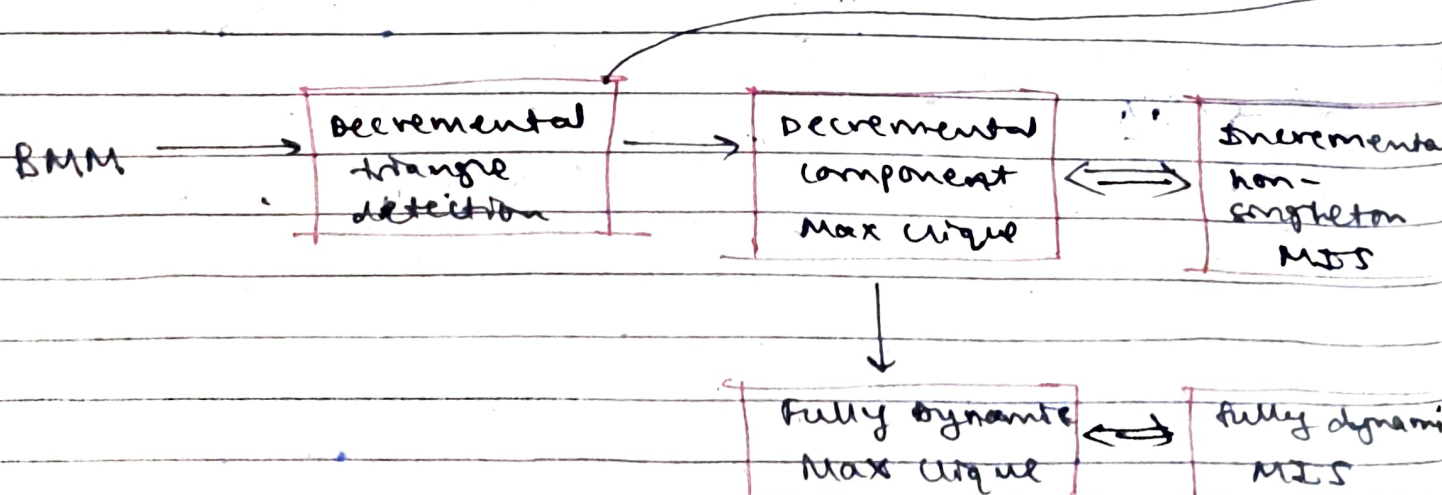
A maximal clique on graph G is the maximal independent set on its complement graph G^c .

proof: If $\exists S \notin M$ such that $M \cup \{s\}$ is the MIS on G^c then there is no edge between (s, u) for $u \in M$ in G^c
 $\Rightarrow$ there are edges b/w all (s, u) for $u \in M$ in $G \Rightarrow s \in M$. Hence max clique on G is MIS on G^c . Similarly, we can show the other direction.

this implies the following two results:

- (i) Decremental max clique $\Leftrightarrow$ incremental MIS
- (ii) Incremental max clique $\Leftrightarrow$ decremental MIS

Hence, the reductions can be extended to:



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we can reduce BMM to incremental MIS. The idea is to construct a copy of G^c . Suppose S is the set of singleton MIS in G^c .

then we join each $s \in S(G_c^{(i)})$ to $t \in S(G_c^{(i+1)})$. this ensures that

whenever we compute MIS on this graph, it will always decompose in a singleton and non-singleton MIS of S or maybe just a non-singleton MIS.

Hence, we repeat the algorithm used for incremental non-singleton MIS and if insertion of edges ~~creates~~ or extends the set S , we add the edges $s \in S$ to $t \in S$ to ensure this is satisfied

Incremental
MIS

$\Leftrightarrow$

Decremental
max clique

Hence:

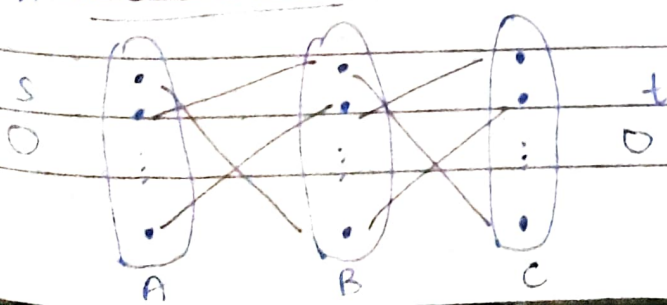
$$P(n) + O(n^2) \quad U(n) + O(n^2) \quad Q(n) = O(n^3)$$

$\Rightarrow$ Incremental MIS cannot be solved in $O(n+E)$

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We can ~~reduce~~ ^{reduce} BMM to fully dynamic s-t reachability

A construction



For each $i=1$ to N :

Add $(S, a(i))$

For each $j=1$ to N :

Add $(c(j), t)$

$M[i][j] = \text{Query}$

Delete $(c(j), t)$

Delete $(S, a(i))$

this implies

$$P(n) = O(n^2) U(n) + O(n^2) Q(n) = O(n^3)$$

and establishes

BMM $\rightarrow$ fully dynamic
s-t reachability

However, we already know:

(i) OMV $\rightarrow$ fully dynamic
s-t reachability

(ii) BMM $\rightarrow$ OMV

So the entire picture is:

